

Software Requirements Specification

for

Vehicle Parking System

A system for managing parking slots and vehicle tracking

Version 1.1 approved

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Technology: MERN Stack (MongoDB, Express.js, React.js, Node.js)

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Revision History

Name	Date	Reason For Changes	Version
Team (Ramitha, Sneetha, Pramithi)	04-10-2026	Initial SRS prepared for Vehicle Parking System	1.0
Team (Ramitha, Sneetha, Pramithi)	04-10-2026	Scope reduced for a three-member full-stack project: Vehicle Owner role removed; 23 FRs, 7 NFRs, 7 security requirements and 5 business rules; measurable requirements; complete RTM; redrawn use case diagram	1.1

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) specifies the requirements of the Vehicle Parking System (VPS), Version 1.1. The VPS is a web-based system for managing parking slots and tracking vehicles in a parking facility. This document covers the complete system and defines its functional requirements, non-functional requirements, security objectives and requirements, interfaces, assumptions and design constraints.

1.2 Intended Audience

This document is intended for the project development team, testers, project reviewers and instructors, parking administrators and attendants, and future maintainers. Section 2 gives the overall description, Section 3 the external interfaces, Section 4 the analysis model (use case diagram), Section 5 the functional requirements grouped by system feature, Section 6 the non-functional, security and business-rule requirements, and Section 7 other requirements. Appendices A to C contain the glossary, field layouts and the requirements traceability matrix.

1.3 Product Scope

The VPS lets a parking facility keep an accurate, current record of which slots are free, which vehicle is parked in which slot, when it entered and left, and how much it owes. Its objectives are to:

- Register vehicle information.
- Show the availability and status of every parking slot.
- Allocate an available slot of the right type to an entering vehicle.
- Record vehicle entry and exit and release the slot on exit.
- Calculate the parking fee from the parking duration and configured fee rules.
- Track vehicles: find a vehicle, show its current location and keep its parking history.
- Let administrators manage slots, users, fee rates and the occupancy report.
- Restrict every protected operation to authorized users.

Out of scope: online payment, number-plate recognition, hardware gate control, reservations and mobile-native applications.

1.4 Definitions, Acronyms, and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
SAD	Software Architecture & Design Specification
VPS	Vehicle Parking System
FR / NFR / SEC / BR	Functional Requirement / Non-Functional Requirement / Security Requirement / Business Rule
RTM	Requirements Traceability Matrix
UI	User Interface
API	Application Programming Interface
REST	Representational State Transfer

Term	Meaning
JSON	JavaScript Object Notation
HTTP / HTTPS	Hypertext Transfer Protocol / HTTP over TLS
TLS	Transport Layer Security
RBAC	Role-Based Access Control
JWT	JSON Web Token
MERN	MongoDB, Express.js, React.js, Node.js
ID	Identifier

1.5 References

1. PES University, Software Engineering Mini-Project Deliverables Part-1, ETA 07-10-2026.
2. PES University, Software Requirements Specification template (IEEE-based), Version 1.0.
3. PES University, Project Guidelines for the Software Engineering mini-project.
4. Project problem statement No. 10: "Vehicle Parking System – A system for managing parking slots and vehicle tracking."
5. IEEE Std 830-1998 / ISO/IEC/IEEE 29148:2018, Systems and software engineering – Requirements engineering.
6. MongoDB, Express.js, React.js and Node.js official documentation.

2. Overall Description

2.1 Product Perspective

The VPS is a new, self-contained web application. A React.js frontend runs in the user's browser and calls a Node.js/Express.js backend through a REST API over HTTP/HTTPS. The backend applies the business rules and stores all data in a MongoDB database. The VPS has no interface to any other system in Version 1.0.

2.2 Product Functions

- Authentication, user management and role-based access control.
- Vehicle registration.
- Parking slot management and slot availability display.
- Vehicle entry recording and automatic slot allocation.
- Vehicle exit recording, parking fee calculation and slot release.
- Parking fee rate configuration.
- Vehicle search, current-location tracking and parking history.
- Parking occupancy report.

2.3 User Classes and Characteristics

User class	Characteristics	Main functions	Importance
Parking Attendant	Facility staff; basic computer skill; uses the system all day at the entry and exit points.	Register vehicles; view slot availability; record entry and exit; allocate slots; search vehicles; view parking history.	High
Administrator	Facility manager; moderate technical skill; highest privileges.	All attendant functions; manage slots, users and fee rates; generate the occupancy report.	High

2.4 Operating Environment

- Frontend: React.js application in the latest two stable versions of Chrome, Firefox or Edge on desktop or mobile.
- Backend: Node.js with Express.js (LTS version) on a server or cloud host.
- Database: MongoDB.
- Communication: HTTP/HTTPS with JSON; network connectivity is required.

2.5 Design and Implementation Constraints

- The system shall be implemented with the MERN stack (the problem statement permits JavaScript/MERN).
- All data shall be stored in a persistent MongoDB database.
- Protected functions shall be available only to authorized users (see Section 6.3).
- The SRS, SAD and Test Plan shall remain traceable to each other through the RTM (Appendix C).
- The system shall be developed by the three-member team within the mini-project schedule.

2.6 Assumptions and Dependencies

Assumptions

- Every slot has a unique Slot ID and a fixed slot type.
- The facility has one entry/exit point operated by attendants who use the web application.
- Hourly rates are set by an Administrator before the system is used. The team proposes these default values, to be confirmed before implementation: Two-wheeler ₹10 per hour, Car ₹30 per hour, Other ₹50 per hour. The grace period is fixed at 15 minutes.
- Fees are collected outside the system (cash or counter); the system only calculates and records them.
- Network connectivity is available whenever the system is used.

Dependencies

- Availability of MongoDB, Node.js and a supported web browser.
- Successful integration of the React.js frontend, Express.js backend and MongoDB database.
- Confirmation of the default hourly rates by the project reviewer or customer.

3. External Interface Requirements

3.1 User Interfaces

The VPS shall provide a responsive web interface with these screens: Login; Slot Availability; Vehicle Registration and Search; Vehicle Entry; Vehicle Exit with fee summary; Parking History; Slot Management; User Management; Fee Rates; Occupancy Report. Navigation items shall be shown only for functions permitted to the logged-in role.

- Mandatory form fields shall be marked with an asterisk, and invalid input shall show a message next to the field.
- Slot status shall be shown with text and colour: Available (green), Occupied (red), Unavailable (grey).
- Every screen shall show the logged-in user, the role and a Logout control.

3.2 Software Interfaces

The React.js frontend shall call the Express.js backend through a REST API with JSON requests and responses. The backend shall access MongoDB through a database driver or ODM. Data exchanged: user credentials and session token; vehicle ID, registration number and type; slot ID, type and status; entry and exit timestamps; duration; fee; report data.

3.3 Communications Interfaces

- The frontend and backend shall communicate over HTTP; the deployed system shall use HTTPS (SEC-04).
- Requests and responses shall use JSON and standard HTTP status codes (200, 201, 400, 401, 403, 404, 409, 500).
- Error responses shall not reveal stack traces or database details.

3.4 Hardware Interfaces

None. The system runs on standard client devices with a web browser and on a server capable of running Node.js and MongoDB.

4. Analysis Models

4.1 Use Case Diagram

Figure 1 shows the actors and use cases of the VPS.

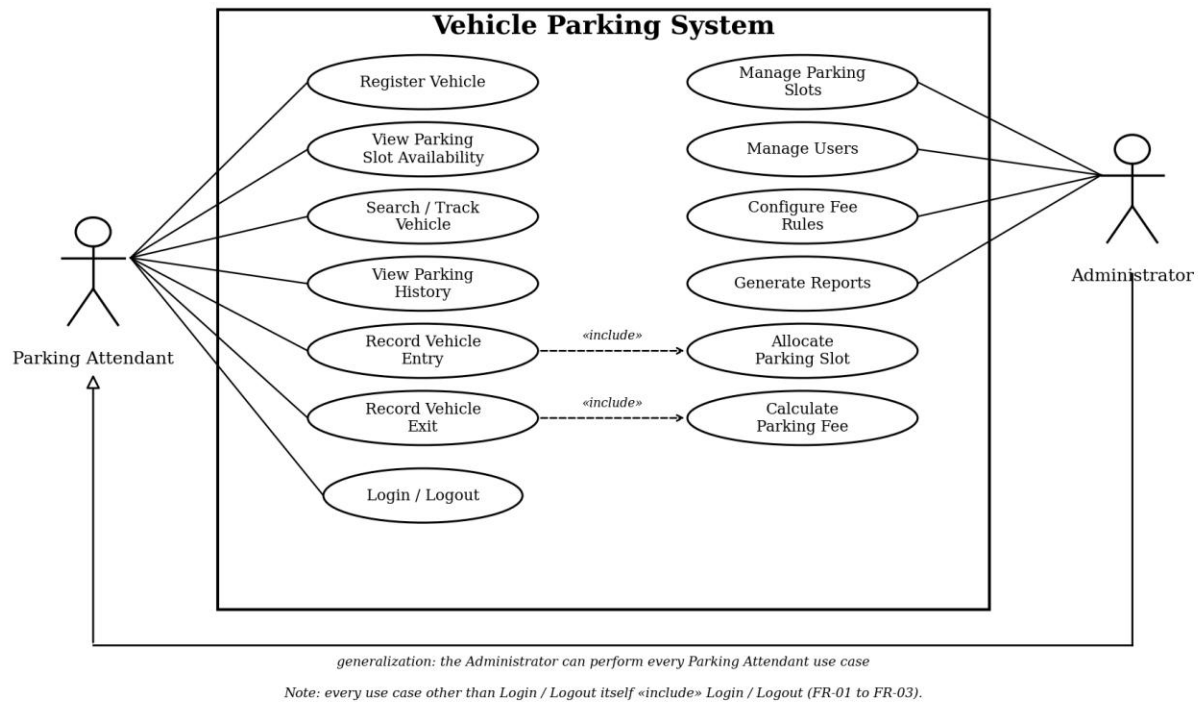


Figure 1: Use case diagram of the Vehicle Parking System

4.2 Use Case Summary

Use case	Actor(s)	Related requirements
Login / Logout	Attendant, Administrator	FR-01 to FR-03, FR-05
Register Vehicle	Attendant, Administrator	FR-06, FR-07
View Parking Slot Availability	Attendant, Administrator	FR-10
Search / Track Vehicle	Attendant, Administrator	FR-20, FR-21
View Parking History	Attendant, Administrator	FR-22
Record Vehicle Entry (includes Allocate Parking Slot)	Attendant, Administrator	FR-11 to FR-14
Record Vehicle Exit (includes Calculate Parking Fee)	Attendant, Administrator	FR-15 to FR-18
Manage Parking Slots	Administrator	FR-08, FR-09
Manage Users	Administrator	FR-04
Configure Fee Rules	Administrator	FR-19
Generate Reports	Administrator	FR-23

5. System Features and Functional Requirements

Each requirement has a unique ID, states one testable behaviour, and uses "shall". Priority is High, Medium or Low.

5.1 Authentication and Access Control

Description and priority: The system shall let users log in with a username and password and restrict every protected function to authenticated users holding the required role (Parking Attendant or Administrator). Priority: **High**.

Stimulus/response sequence: The user submits credentials → the system validates them → on success the system opens a session and shows the role-specific home page; on failure the system shows a generic error.

Functional requirements:

FR-01: The system shall authenticate a user by username and password and, on success, issue a session token that identifies the user and role.

FR-02: The system shall reject invalid credentials with the single message "Invalid username or password", without indicating which field was wrong.

FR-03: The system shall allow a logged-in user to log out, after which the session token shall no longer be accepted.

FR-04: The system shall allow an Administrator to create, update and deactivate Parking Attendant and Administrator accounts and to assign a role to each account.

FR-05: The system shall deny a request for a function not permitted to the requester's role with HTTP status 403 and a message that contains no protected data.

5.2 Vehicle Registration

Description and priority: The system shall maintain a record of every vehicle that uses the parking facility. Priority: **High**.

Stimulus/response sequence: The attendant enters registration number and vehicle type → the system validates the input → the system stores the vehicle → the system confirms the registration.

Functional requirements:

FR-06: The system shall allow a Parking Attendant or Administrator to register a vehicle by entering a unique registration number, a vehicle type (Two-wheeler, Car or Other) and an optional owner contact number; registration numbers of 4 to 15 alphanumeric characters (hyphens and spaces allowed) shall be stored in upper case.

FR-07: The system shall reject a vehicle registration when a mandatory field is empty, the registration number does not meet the format of FR-06, or the registration number already exists, and shall name the offending field in the error message.

5.3 Parking Slot Management

Description and priority: The system shall maintain the parking slots of the facility and show their current status. Priority: **High**.

Stimulus/response sequence: The Administrator enters slot details → the system validates and stores them; any authenticated user opens the slot view → the system displays every slot with its current status.

Functional requirements:

FR-08: The system shall allow an Administrator to add a parking slot with a unique Slot ID and a slot type (Two-wheeler, Car or Other); the initial status shall be Available.

FR-09: The system shall allow an Administrator to set a slot to Unavailable and back to Available, and shall reject this action when the slot is Occupied.

FR-10: The system shall display every parking slot with its Slot ID, slot type and status (Available, Occupied or Unavailable), together with the number of Available and Occupied slots for each slot type, to every authenticated user.

5.4 Vehicle Entry and Slot Allocation

Description and priority: The system shall record the entry of a vehicle and allocate an available slot to it. Priority: **High**.

Stimulus/response sequence: The attendant selects a registered vehicle and records entry → the system checks for an active record and a free slot of the matching type → the system assigns a slot, stores the entry time and sets the slot to Occupied.

Functional requirements:

FR-11: The system shall allow a Parking Attendant or Administrator to record the entry of a registered vehicle, and shall reject the entry when the vehicle already has an active parking record.

FR-12: The system shall automatically allocate the Available slot with the lowest Slot ID whose slot type matches the vehicle type, and shall allow the attendant to choose a different Available slot of the same type.

FR-13: The system shall refuse the entry and display "No parking slot available for this vehicle type" when no matching Available slot exists.

FR-14: The system shall store the entry time (server time), the allocated Slot ID and a unique ticket number in a parking record and set the slot to Occupied as a single atomic operation.

5.5 Vehicle Exit and Fee Calculation

Description and priority: The system shall record the exit of a vehicle, calculate the parking fee and release the slot. Priority: **High**.

Stimulus/response sequence: The attendant searches the vehicle and records exit → the system stores the exit time and calculates the fee → the attendant confirms → the record is closed and the slot is set to Available.

Functional requirements:

FR-15: The system shall allow a Parking Attendant or Administrator to record the exit of a vehicle and shall store the exit time (server time) in its active parking record; the system shall reject the exit when the vehicle has no active parking record.

FR-16: The system shall calculate the fee as the number of billable hours multiplied by the hourly rate of the vehicle type, where billable hours is the parking duration after subtracting a fixed grace period of 15 minutes, rounded up to the next whole hour, and the fee is 0 when the duration does not exceed 15 minutes.

FR-17: The system shall display the calculated fee, entry time, exit time and duration to the attendant before the exit is confirmed.

FR-18: The system shall, on confirmation of the exit, mark the parking record Completed with the fee, and set the slot status to Available.

5.6 Parking Fee Rule Configuration

Description and priority: The system shall let the Administrator define the hourly rates used to calculate parking fees. Priority: **Medium**.

Stimulus/response sequence: The Administrator enters a rate for a vehicle type → the system validates the value → the system stores it for use in later fee calculations.

Functional requirements:

FR-19: The system shall allow an Administrator to set the hourly rate, in rupees, for each vehicle type; the rate shall be a number greater than or equal to 0, and a changed rate shall apply only to exits recorded after the change.

5.7 Vehicle Search, Tracking and History

Description and priority: The system shall let users find a vehicle, see where it is parked now and review its past parking records. Priority: **High**.

Stimulus/response sequence: The user enters a registration number → the system looks up the vehicle → the system shows its current parking status and its parking history.

Functional requirements:

FR-20: The system shall allow a Parking Attendant or Administrator to search a registered vehicle by registration number.

FR-21: The system shall display the current status of a searched vehicle as Parked (with Slot ID, entry time and elapsed duration) or Not Parked.

FR-22: The system shall display the parking history of a vehicle, showing for each past record the entry time, exit time, Slot ID and fee, ordered with the most recent first.

5.8 Reports

Description and priority: The system shall let the Administrator view the occupancy of the facility. Priority: **Medium**.

Stimulus/response sequence: The Administrator selects the occupancy report → the system queries the slots and active records → the system displays the report.

Functional requirements:

FR-23: The system shall allow an Administrator to generate a Parking Occupancy Report listing each slot with its status, assigned vehicle and last updated time.

6. Other Non-Functional Requirements

6.1 Performance Requirements

NFR-01: Search for a vehicle (FR-20) shall return its result within 3 seconds for at least 95% of requests with up to 50 concurrent users.

NFR-02: Recording a vehicle entry including slot allocation (FR-11 to FR-14) shall complete within 3 seconds for at least 95% of operations.

NFR-03: Recording a vehicle exit including fee calculation (FR-15 to FR-17) shall complete within 3 seconds for at least 95% of operations.

6.2 Safety and Reliability Requirements

NFR-04: When two entry requests try to allocate the same slot at the same time, exactly one shall succeed and the other shall be told to retry; a slot shall never have two active vehicles.

NFR-05: Every entry or exit operation acknowledged as successful shall be persisted in the database; no acknowledged record shall be lost after a server restart.

6.3 Security Requirements

Security objectives

- SO-1: Confidentiality – protect user, vehicle and parking data from unauthorized access.
- SO-2: Integrity – keep slot status, parking records and fee rules correct and change them only through authorized, validated operations.
- SO-3: Access control – allow privileged operations only to users with the required role.

Security requirements

SEC-01: The system shall authenticate every user before granting access to any function other than login.

SEC-02: The system shall enforce role-based authorization on the server for every API endpoint, using the roles Parking Attendant and Administrator.

SEC-03: The system shall store passwords only as salted bcrypt hashes with a cost factor of at least 10, and shall never write passwords to logs.

SEC-04: The deployed system shall transmit all client-server traffic over HTTPS using TLS 1.2 or higher.

SEC-05: The system shall validate the type, length and format of every input on the server and shall reject input containing database operators or script content.

SEC-06: The system shall end a session after 15 minutes of inactivity.

SEC-07: Database credentials and token-signing secrets shall be kept in environment configuration and not in source code.

6.4 Software Quality Attributes

NFR-06: Usability: a trained Parking Attendant shall be able to record a vehicle entry in no more than 5 user actions and within 60 seconds.

NFR-07: Portability: the web interface shall work on the latest two stable versions of Chrome, Firefox and Edge and on screen widths from 360 px upward.

6.5 Business Rules

BR-01: A parking slot shall have at most one active parking record at a time.

BR-02: Only slots with status Available and a slot type matching the vehicle type shall be allocated.

BR-03: A vehicle shall have at most one active parking record at a time, and a vehicle exit shall require an active parking record.

BR-04: Parking fees shall be calculated only from the hourly rates configured by an Administrator and the fixed grace period of 15 minutes.

BR-05: Only an Administrator shall manage slots, users, fee rates and reports.

7. Other Requirements

- Database: vehicle, slot, user, parking-record and fee-rate data shall be kept in separate MongoDB collections with unique indexes on registration number, Slot ID and username.
- Time: all timestamps shall be recorded in server time (IST) and stored in UTC.
- Currency: all amounts shall be in Indian rupees (₹) with two decimal places.
- Traceability: the RTM in Appendix C shall be kept up to date as the SAD, code and Test Plan are produced.
- Open items: confirmation of default hourly rates (Section 2.6) and the group number on the cover page.

Appendix A: Glossary

Term	Definition
Vehicle	A two-wheeler, car or other vehicle registered in the system.
Parking Slot	A uniquely identified parking space with a slot type and a status.
Slot Status	Available (free), Occupied (has an active vehicle) or Unavailable (out of service).
Parking Record	A record of one visit: vehicle, slot, entry time, exit time, fee and status (Active or Completed).
Vehicle Tracking	Finding a vehicle and showing its current parking status and past parking records.
Grace Period	The first 15 minutes of a visit, for which no fee is charged (fixed).
Parking Fee	Charge calculated from billable hours and the hourly rate of the vehicle type.
Parking Attendant	Staff member who records vehicle entry and exit.
Administrator	Authorized user who manages slots, users, fee rates and the occupancy report.

Appendix B: Field Layouts

Proposed layouts for Version 1.1; lengths and types may be refined in detailed design.

B.1 Vehicle

Field	Length	Data Type	Description	Mandatory
Vehicle ID	24	Alphanumeric	System-generated unique identifier	Y
Registration Number	15	Alphanumeric	Number plate, unique, upper case	Y
Vehicle Type	20	String	Two-wheeler, Car or Other	Y
Owner Contact	15	Numeric	Owner phone number	N
Active	1	Boolean	Vehicle record in use	Y

B.2 Parking Slot

Field	Length	Data Type	Description	Mandatory
Slot ID	12	Alphanumeric	Unique slot identifier, e.g. A-01	Y
Slot Type	20	String	Two-wheeler, Car or Other	Y

Field	Length	Data Type	Description	Mandatory
Status	15	String	Available, Occupied or Unavailable	Y
Last Updated	20	Date-Time	Time of last status change	Y

B.3 Parking Record

Field	Length	Data Type	Description	Mandatory
Ticket Number	16	Alphanumeric	Unique number for the visit	Y
Vehicle ID	24	Alphanumeric	Reference to the vehicle	Y
Slot ID	12	Alphanumeric	Slot assigned	Y
Entry Time	20	Date-Time	Server time of entry	Y
Exit Time	20	Date-Time	Server time of exit	N
Status	15	String	Active or Completed	Y
Parking Fee	12	Decimal (2 places)	Fee in ₹, set on completion	N

B.4 User

Field	Length	Data Type	Description	Mandatory
User ID	24	Alphanumeric	System-generated identifier	Y
Username	30	Alphanumeric	Unique login name	Y
Name	60	String	Full name	Y
Password Hash	60	String	bcrypt hash (never the password)	Y
Role	20	String	Attendant or Administrator	Y
Active	1	Boolean	Account enabled	Y

B.5 Fee Rate

Field	Length	Data Type	Description	Mandatory
Vehicle Type	20	String	Type the rate applies to	Y
Hourly Rate	10	Decimal (2 places)	Rate in ₹ per hour	Y

B.6 Parking Occupancy Report Fields

Field	Description
Slot ID	Slot identifier
Slot Type	Two-wheeler, Car or Other
Slot Status	Available, Occupied or Unavailable
Assigned Vehicle	Registration number of the parked vehicle (blank if none)
Last Updated Time	Time of last status change

Appendix C: Requirement Traceability Matrix

The matrix links every requirement in this SRS to its architecture reference, design reference, code file and test cases. Architecture, design, code and test case references are filled in as the SAD, implementation and Test Plan are produced; "TBD" marks references not yet assigned.

Sl. No	Req. ID	Brief Description	Architecture Reference	Design Reference	Code File Ref.	Test Case ID
1	FR-01	The system shall authenticate a user by username and password and, on success, issue a session token that identifies the user and role.	Auth Service	Login API Design	TBD	TC-AUTH-01
2	FR-02	The system shall reject invalid credentials with the single message "Invalid username or password", without indicating which field was wrong.	Auth Service	Login API Design	TBD	TC-AUTH-02
3	FR-03	The system shall allow a logged-in user to log out, after which the session token shall no longer be accepted.	Auth Service	Logout / Session Design	TBD	TC-AUTH-03
4	FR-04	The system shall allow an Administrator to create, update and deactivate Parking Attendant and Administrator accounts and to assign a role to each account.	User Management Service	User Management Design	TBD	TC-AUTH-04
5	FR-05	The system shall deny a request for a function not permitted to the requester's role with HTTP status 403 and a message that contains no protected data.	Security Middleware	Authorization Design	TBD	TC-AUTH-05
6	FR-06	The system shall allow a Parking Attendant or Administrator to register a vehicle by entering a unique registration number, a vehicle type (Two-wheeler, Car or Other) and an optional owner contact number; registration numbers of 4 to 15 alphanumeric characters (hyphens and spaces allowed) shall be stored in upper case.	Vehicle Service	Vehicle Registration Design	TBD	TC-VEH-01, TC-VEH-03
7	FR-07	The system shall reject a vehicle registration when a mandatory field is empty, the registration number does not meet the format of FR-06, or the registration number already exists, and shall name the offending field in the error message.	Vehicle Service	Input Validation Design	TBD	TC-VEH-02, TC-VEH-03
8	FR-08	The system shall allow an Administrator to add a parking slot with a unique Slot ID and a slot type (Two-wheeler, Car or Other); the initial status shall be Available.	Parking Slot Service	Slot Management Design	TBD	TC-SLOT-01
9	FR-09	The system shall allow an Administrator to set a slot to Unavailable and back to Available, and shall reject this action when the slot is Occupied.	Parking Slot Service	Slot Management Design	TBD	TC-SLOT-02

Sl. No	Req. ID	Brief Description	Architecture Reference	Design Reference	Code File Ref.	Test Case ID
10	FR-10	The system shall display every parking slot with its Slot ID, slot type and status (Available, Occupied or Unavailable), together with the number of Available and Occupied slots for each slot type, to every authenticated user.	Parking Slot Service	Slot Availability Design	TBD	TC-SLOT-03
11	FR-11	The system shall allow a Parking Attendant or Administrator to record the entry of a registered vehicle, and shall reject the entry when the vehicle already has an active parking record.	Parking Transaction Service	Entry Flow Design	TBD	TC-ENT-01, TC-ENT-02
12	FR-12	The system shall automatically allocate the Available slot with the lowest Slot ID whose slot type matches the vehicle type, and shall allow the attendant to choose a different Available slot of the same type.	Parking Transaction Service	Slot Allocation Design	TBD	TC-ENT-01, TC-ENT-04
13	FR-13	The system shall refuse the entry and display "No parking slot available for this vehicle type" when no matching Available slot exists.	Parking Transaction Service	Slot Allocation Design	TBD	TC-ENT-03
14	FR-14	The system shall store the entry time (server time), the allocated Slot ID and a unique ticket number in a parking record and set the slot to Occupied as a single atomic operation.	Parking Transaction Service	Entry Flow Design	TBD	TC-ENT-01
15	FR-15	The system shall allow a Parking Attendant or Administrator to record the exit of a vehicle and shall store the exit time (server time) in its active parking record; the system shall reject the exit when the vehicle has no active parking record.	Parking Transaction Service	Exit Flow Design	TBD	TC-EXT-01, TC-EXT-02
16	FR-16	The system shall calculate the fee as the number of billable hours multiplied by the hourly rate of the vehicle type, where billable hours is the parking duration after subtracting a fixed grace period of 15 minutes, rounded up to the next whole hour, and the fee is 0 when the duration does not exceed 15 minutes.	Fee Service	Fee Calculation Design	TBD	TC-EXT-03, TC-FEE-02
17	FR-17	The system shall display the calculated fee, entry time, exit time and duration to the attendant before the exit is confirmed.	Fee Service	Exit Flow Design	TBD	TC-EXT-01
18	FR-18	The system shall, on confirmation of the exit, mark the parking record Completed with the fee, and set the slot status to Available.	Parking Transaction Service	Slot Release Design	TBD	TC-EXT-01
19	FR-19	The system shall allow an Administrator to set the hourly rate, in rupees, for each vehicle type; the rate shall be a number greater than or equal to 0, and a changed rate shall apply only to exits recorded after the change.	Fee Service	Fee Rule Design	TBD	TC-FEE-01, TC-FEE-02

Sl. No	Req. ID	Brief Description	Architecture Reference	Design Reference	Code File Ref.	Test Case ID
20	FR-20	The system shall allow a Parking Attendant or Administrator to search a registered vehicle by registration number.	Vehicle Service	Vehicle Search Design	TBD	TC-SRCH-01
21	FR-21	The system shall display the current status of a searched vehicle as Parked (with Slot ID, entry time and elapsed duration) or Not Parked.	Vehicle Service / Parking Transaction Service	Vehicle Tracking Design	TBD	TC-SRCH-02
22	FR-22	The system shall display the parking history of a vehicle, showing for each past record the entry time, exit time, Slot ID and fee, ordered with the most recent first.	Parking Transaction Service	Parking History Design	TBD	TC-SRCH-03
23	FR-23	The system shall allow an Administrator to generate a Parking Occupancy Report listing each slot with its status, assigned vehicle and last updated time.	Report Service	Report Design	TBD	TC-REP-01
24	NFR-01	Search for a vehicle (FR-20) shall return its result within 3 seconds for at least 95% of requests with up to 50 concurrent users.	Application/API Layer	Performance Design	TBD	TC-PERF-01
25	NFR-02	Recording a vehicle entry including slot allocation (FR-11 to FR-14) shall complete within 3 seconds for at least 95% of operations.	Parking Transaction Service	Performance Design	TBD	TC-PERF-02
26	NFR-03	Recording a vehicle exit including fee calculation (FR-15 to FR-17) shall complete within 3 seconds for at least 95% of operations.	Parking Transaction Service	Performance Design	TBD	TC-PERF-02
27	NFR-04	When two entry requests try to allocate the same slot at the same time, exactly one shall succeed and the other shall be told to retry; a slot shall never have two active vehicles.	Parking Transaction Service	Concurrency Control Design	TBD	TC-CONC-01
28	NFR-05	Every entry or exit operation acknowledged as successful shall be persisted in the database; no acknowledged record shall be lost after a server restart.	Database Layer	Data Persistence Design	TBD	TC-REL-01
29	NFR-06	Usability: a trained Parking Attendant shall be able to record a vehicle entry in no more than 5 user actions and within 60 seconds.	Presentation Layer	UI Design	TBD	TC-USAB-01
30	NFR-07	Portability: the web interface shall work on the latest two stable versions of Chrome, Firefox and Edge and on screen widths from 360 px upward.	Presentation Layer	UI Design	TBD	TC-PORT-01
31	SEC-01	The system shall authenticate every user before granting access to any function other than login.	Security Middleware	Authentication Design	TBD	TC-AUTH-01, TC-SEC-01
32	SEC-02	The system shall enforce role-based authorization on the server for every API endpoint, using the roles Parking Attendant and Administrator.	Security Middleware	Authorization Design	TBD	TC-AUTH-05, TC-SEC-02

Sl. No	Req. ID	Brief Description	Architecture Reference	Design Reference	Code File Ref.	Test Case ID
33	SEC-03	The system shall store passwords only as salted bcrypt hashes with a cost factor of at least 10, and shall never write passwords to logs.	Auth Service	Password Storage Design	TBD	TC-SEC-03
34	SEC-04	The deployed system shall transmit all client-server traffic over HTTPS using TLS 1.2 or higher.	Deployment Architecture	Secure Communication Design	TBD	TC-SEC-04
35	SEC-05	The system shall validate the type, length and format of every input on the server and shall reject input containing database operators or script content.	Security Middleware	Input Validation Design	TBD	TC-SEC-05
36	SEC-06	The system shall end a session after 15 minutes of inactivity.	Auth Service	Session Design	TBD	TC-SEC-06
37	SEC-07	Database credentials and token-signing secrets shall be kept in environment configuration and not in source code.	Deployment Architecture	Configuration Design	TBD	TC-SEC-07

Business rules BR-01 to BR-05 are enforced through FR-11 to FR-19 and NFR-04 and are verified by the same test cases.